

Technology #35

increase the sensing threshold of the
leadless pacemaker to suppress oversensing

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

increase the sensing threshold of the leadless pacemaker to suppress oversensing

WHY DOES IT MATTER?

Pacemaker battery depletion requiring surgical replacement

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 12 design inputs, 14 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in rf_wireless. 6-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

canonical invention specification (with epistemic classes); engineering specification (this document, all blocks classed); design input/output graph with explicit IDs; engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification); killer-experiment definition and its EIG basis; evidence custody references and hashes; unresolved-uncertainty register; domain equation set (symbolic; sources and applicability judgments identified)

WHAT DOES THE BUYER HAVE TO BUILD?

actual geometry and detailed design; sourced engineering parameters and acceptance thresholds; physical prototypes; all verification and validation results; manufacturing process qualification; regulatory pathway determination and submissions

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (pre-registered pass/fail required before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 2 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 6-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: record pacing capture and oversensing events before and after threshold adjustment; confirm that oversensing events cease and pacing inhi.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: link-budget verification in tissue-equivalent phantom. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: in-body path loss exceeds free-space — margin is MODEL_DERIVED; noise figure and interference add on top of the thermal floor and must be budgeted explicitly; noise figure and interference terms must be budgeted explicitly. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

2 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: mechanically mediated device-device interaction causing intermittent oversensing

POTENTIAL VALUE DRIVER: Addresses: Pacemaker battery depletion requiring surgical replacement

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: rf_wireless

Disciplines: RF engineering, antenna design, signal processing, EMC

Architecture: PROPOSED architecture (no physical system exists): transmit chain + power limits; channel / tissue path (path loss model); receive chain + detection; regulatory exposure constraint (SAR) — realizing: increase the sensing threshold of the leadless pacemaker to suppress oversensing

Subsystems:

?: transmit chain + power limits - ?

?: channel / tissue path (path loss model) - ?

?: receive chain + detection - ?

?: regulatory exposure constraint (SAR) - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: increase the sensing threshold of the leadless pacemaker to suppress oversensing

Key Physics: RF / wireless link governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: RF / wireless link governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

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{'equation_id': 'RF-001', 'name': 'Friis link budget', 'expression': 'P_r = P_t * G_t * G_r * (lambda / (4 * pi * d))^2', 'source': {'text': 'Friis transmission equation (standard antenna-theory relation)', 'epistemic_class': 'EXTERNAL_PRECEDENT', 'verify_before_release': True}, 'applicability': {'condition': 'far field, line of sight; tissue path requires additional loss term', 'epistemic_class': 'MODEL_DERIVED', 'judged_for_domain': 'rf_wireless'}, 'assumptions': ['in-body path loss exceeds free-space – margin is MODEL_DERIVED'], 'selection_rationale': {'epistemic_class': 'MODEL_DERIVED', 'verdict': 'CONDITIONAL', 'reason': 'domain template justifies this relation for the domain, but the invention text does not yet engage its variables; applicability must be confirmed by the first characterization', 'engaged_variables': [], 'assumption_check': {'assumptions': ['in-body path loss exceeds free-space – margin is MODEL_DERIVED'], 'violations': [], 'epistemic_class': 'MODEL_DERIVED (mechanically evaluated, recorded for audit)'}, 'mechanism_mentions': [], 'model_applicability': 'far field, line of sight; tissue path requires additional loss term', 'model_assumptions': ['in-body path loss exceeds free-space – margin is MODEL_DERIVED'], 'invention_tie': {'linkage_kind': 'applicability_judgment', 'verdict': 'CONDITIONAL', 'engaged_variables': [], 'mechanism_mention_count': 0}}
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{'equation_id': 'RF-003', 'name': 'Thermal noise floor', 'expression': 'N = k * T * B', 'source': {'text': 'standard thermal-noise relation (communications texts)', 'epistemic_class': 'EXTERNAL_PRECEDENT', 'verify_before_release': True}, 'applicability': {'condition': 'matched, linear receiver at physical temperature T', 'epistemic_class': 'MODEL_DERIVED', 'judged_for_domain': 'rf_wireless'}, 'assumptions': ['noise figure and interference add on top of the thermal floor and must be budgeted explicitly'], 'selection_rationale': {'epistemic_class': 'MODEL_DERIVED', 'verdict': 'CONDITIONAL', 'reason': 'domain template justifies this relation for the domain, but the invention text does not yet engage its variables; applicability must be confirmed by the first characterization', 'engaged_variables': [], 'assumption_check': {'assumptions': ['noise figure and interference add on top of the thermal floor and must be budgeted explicitly'], 'violations': [], 'epistemic_class': 'MODEL_DERIVED (mechanically evaluated, recorded for audit)'}, 'mechanism_mentions': [], 'model_applicability': 'matched, linear receiver at physical temperature T', 'model_assumptions': ['noise figure and interference add on top of the thermal floor and must be budgeted explicitly'], 'invention_tie': {'linkage_kind': 'applicability_judgment', 'verdict': 'CONDITIONAL', 'engaged_variables': [], 'mechanism_mention_count': 0}}
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{'equation_id': 'RF-002', 'name': 'Signal-to-noise ratio', 'expression': 'SNR = P_signal / P_noise', 'source': {'text': 'standard detection-theory quantity (communications texts)', 'epistemic_class': 'EXTERNAL_PRECEDENT', 'verify_before_release': True}, 'applicability': {'condition': 'defined for the
```

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chosen detection bandwidth', 'epistemic_class': 'MODEL_DERIVED', 'judged_for_domain': 'rf_wireless'},
'assumptions': ['noise figure and interference terms must be budgeted explicitly'], 'selection_rationale':
{'epistemic_class': 'MODEL_DERIVED', 'verdict': 'CONDITIONAL', 'reason': 'domain template justifies this
relation for the domain, but the invention text does not yet engage its variables; applicability must be
confirmed by the first characterization', 'engaged_variables': [], 'assumption_check': {'assumptions':
['noise figure and interference terms must be budgeted explicitly'], 'violations': [], 'epistemic_class':
'MODEL_DERIVED (mechanically evaluated, recorded for audit)'}, 'mechanism_mentions': []},
'model_applicability': 'defined for the chosen detection bandwidth', 'model_assumptions': ['noise figure
and interference terms must be budgeted explicitly'], 'invention_tie': {'linkage_kind':
'applicability_judgment', 'verdict': 'CONDITIONAL', 'engaged_variables': [], 'mechanism_mention_count': 0}}
```

Assumptions:

- in-body path loss exceeds free-space — margin is MODEL_DERIVED
- noise figure and interference add on top of the thermal floor and must be budgeted explicitly
- noise figure and interference terms must be budgeted explicitly

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'BATTERY_FAILURE', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'adversarial dimension: unsupported_mechanism', 'basis': 'COMPUTED (attack engine verdict)'}
- {'regime': 'adversarial dimension: weak_transfer', 'basis': 'COMPUTED (attack engine verdict)'}
- {'regime': 'adversarial dimension: obvious_combination', 'basis': 'COMPUTED (attack engine verdict)'}
- {'regime': 'adversarial dimension: prior_art', 'basis': 'COMPUTED (attack engine verdict)'}
- {'regime': 'adversarial dimension: contradiction', 'basis': 'COMPUTED (attack engine verdict)'}
- {'regime': 'adversarial dimension: boundary_failure', 'basis': 'COMPUTED (attack engine verdict)'}
- {'regime': 'adversarial dimension: engineering_infeasibility', 'basis': 'COMPUTED (attack engine verdict)'}
- {'regime': 'adversarial dimension: regulatory_incompatibility', 'basis': 'COMPUTED (attack engine verdict)'}
- {'regime': 'link margin collapse at maximum depth/orientation', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'SAR exceedance at transmit power needed for link', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'detuning from dielectric loading of nearby tissue', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'interference in clinic RF environment', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'antenna feed joint fatigue failure', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	Must maintain cardiac pacing for 5-10 years without surgical replacement	SOURCE_FACT
DI-002	Mechanism driver (proposed)	mechanically mediated device-device interaction causing intermittent oversensing	MODELLED
DI-003	Expected effect target	reduce pacing inhibition, preserve intrinsic pacing, and extend battery life to 5-10 years without surgical replacement	MODELLED
DI-004	Domain design input pattern: received data applicability NOT ESTABLISHED (candidate pattern; confirm against evidence)	received data applicability NOT ESTABLISHED (candidate pattern; confirm against evidence)	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: implanted data applicability NOT ESTABLISHED (candidate pattern; confirm against evidence)	implanted data applicability NOT ESTABLISHED (candidate pattern; confirm against evidence)	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: received data applicability NOT ESTABLISHED (candidate pattern; confirm against evidence)	received data applicability NOT ESTABLISHED (candidate pattern; confirm against evidence)	ENGINEERING_PROPOSED
DI-007	Domain design input pattern: battery power applicability: TAFE5mtdtjens [battery]	battery power applicability: TAFE5mtdtjens [battery]	ENGINEERING_PROPOSED
DI-008	Custodied observation: Leadless Versense Transcatheter Dual-Channel Pacemaker	https://www.accessdata.fda.gov/drugsatfda_docs/cls2/CTA000011Orig1s001.pdf	3872780ab405907a
DI-009	Custodied observation: Era of Synchrotron X-ray Imaging in Cardiac Pacing	https://www.sciencedirect.com/science/article/pii/S0969806117300096	acadd33a50b6ecb80e61f6
DI-010	Custodied observation: The first case of Right Bundle Branch Block in a Patient with a Leadless Pacemaker	https://www.sciencedirect.com/science/article/pii/S0969806117300096	97566e4eea9ef85c8
DI-011	Custodied observation: When the Pacemaker Goes Wrong: A Case of Pacemaker-Induced Bradycardia	https://www.sciencedirect.com/science/article/pii/S0969806117300096	f525454f01819b2b4d23fc63
DI-012	Custodied observation: Mechanical Coupling of the Pacemaker to the Heart	https://www.sciencedirect.com/science/article/pii/S0969806117300096	7c6725a5e85f0ad69df03

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	transmit chain + power limits realizing increase the sensing threshold of the leadless pacemaker to suppress oversensing	ONLY OFFICIAL	desired architecture, suppression
DO-002	channel / tissue path (path loss model) realizing increase the sensing threshold of the leadless pacemaker to suppress oversensing	SOME SETTING	desired architecture, suppression
DO-003	receive chain + detection realizing increase the sensing threshold of the leadless pacemaker to suppress oversensing	ONLY OFFICIAL	desired architecture, suppression
DO-004	regulatory exposure constraint (SAR) realizing increase the sensing threshold of the leadless pacemaker to suppress oversensing	ONLY OFFICIAL	desired architecture, suppression
DO-005	transmit chain + power limits interfaces with channel / tissue path (path loss model) realizing increase the sensing threshold of the leadless pacemaker to suppress oversensing	ONLY OFFICIAL	desired architecture, suppression
DO-006	channel / tissue path (path loss model) interfaces with receive chain + detection realizing increase the sensing threshold of the leadless pacemaker to suppress oversensing	ONLY OFFICIAL	desired architecture, suppression
DO-007	receive chain + detection interfaces with regulatory exposure constraint (SAR) realizing increase the sensing threshold of the leadless pacemaker to suppress oversensing	ONLY OFFICIAL	desired architecture, suppression
DO-008	Device boundaries: patient/fluid inlet and outlet terminal control/data points required by device, the sensing threshold of the leadless pacemaker	ONLY OFFICIAL	desired architecture, suppression
DO-009	Flow/force/signal path geometry realizing the mechanism of the sensing threshold of the leadless pacemaker to suppress oversensing	ONLY OFFICIAL	desired architecture, suppression
DO-010	Operating range for carrier frequency within which the mechanism holds its state behavior computed bounds for carrier frequency; verification of range	UNKNOWN	desired architecture, suppression
DO-011	Operating range for transmit power (regulatory-bounded) within which the mechanism holds its state behavior computed bounds for transmit power (regulatory-bounded); v	UNKNOWN	desired architecture, suppression
DO-012	Operating range for antenna gain and pattern within which the mechanism holds its state behavior computed bounds for antenna gain and pattern; verification o	UNKNOWN	desired architecture, suppression
DO-013	Operating range for path length and depth within which the mechanism holds its state behavior computed bounds for path length and depth; verification of ra	UNKNOWN	desired architecture, suppression
DO-014	Operating range for receiver noise figure within which the mechanism holds its state behavior computed bounds for receiver noise figure; verification of ran	UNKNOWN	desired architecture, suppression
DO-015	Operating range for required BER at data rate within which the mechanism holds its state behavior computed bounds for required BER at data rate; verification o	UNKNOWN	desired architecture, suppression
DO-016	Sensing -> decision -> actuation loop maintaining the mechanism of the sensing threshold of the leadless pacemaker to suppress oversensing	PROPOSED	desired architecture, suppression
DO-017	Information path: sensing element -> decision/monitor control/data points required by device, the sensing threshold of the leadless pacemaker	ONLY OFFICIAL	desired architecture, suppression
DO-018	Fixture enabling: link-budget verification in tissue-equivalent phantom — rig concept named, built, specification UNKNOWN; acceptance instrumentation	UNKNOWN	desired architecture, suppression
DO-019	Fixture enabling: SAR simulation + measurement against tissue-equivalent phantom — rig concept named, built, specification UNKNOWN; acceptance instrumentation	UNKNOWN	desired architecture, suppression
DO-020	Fixture enabling: bit-error-rate vs distance/depth curve within which the mechanism holds its state behavior computed bounds for bit-error-rate vs distance/depth curve; verification of	UNKNOWN	desired architecture, suppression

ID	Description	Status	Missing Inputs
DO-021	Fixture enabling: tissue-equivalent phantom link characterization	UNKNOWN - rig concept refined, goals specified	Unknown; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
carrier frequency	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
transmit power (regulatory limit)	UNKNOWN (no sourced value)	W	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
antenna gain and pattern	UNKNOWN (no sourced value)	-	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
path length and depth	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
receiver noise figure	UNKNOWN (no sourced value)	W	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
required BER at data rate	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
SAR at tissue interface	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
BATTERY_FAILURE		design control NOT ESTABLISHED	VS-HED2 (requires design work)	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension: unsupported_mechanism		design control NOT ESTABLISHED	VS-HED2 (requires design work)	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension: weak_transfer		design control NOT ESTABLISHED	VS-HED2 (requires design work)	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension: obvious_combination		design control NOT ESTABLISHED	VS-HED2 (requires design work)	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension: prior_art		design control NOT ESTABLISHED	VS-HED2 (requires design work)	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension: contradiction		design control NOT ESTABLISHED	VS-HED2 (requires design work)	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension: boundary_failure		design control NOT ESTABLISHED	VS-HED2 (requires design work)	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension: engineering_infeasibility		design control NOT ESTABLISHED	VS-HED2 (requires design work)	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension: regulatory_incompatibility		design control NOT ESTABLISHED	VS-HED2 (requires design work)	UNKNOWN — untested candidate failure mode (Art. 2)
link margin collapse at maximum depth/orientation	antenna geometry and pattern	NOT LINKED	Link budget with known antenna unless	UNKNOWN — untested candidate failure mode (Art. 2)
SAR exceedance at transmit power needed for link	transmit power schedule and duty cycle	NOT LINKED	Exposure-compliance with known antenna unless	UNKNOWN — untested candidate failure mode (Art. 2)
detuning from dielectric loading of nearby tissue	loading-tolerant antenna topology	NOT LINKED	Characterization across known range	UNKNOWN — untested candidate failure mode (Art. 2)
interference in clinic RF environment	channel selection, filtering, and coding	NOT LINKED	Interference-in-noise budget	UNKNOWN — untested candidate failure mode (Art. 2)
antenna feed joint fatigue failure	strain relief at feed; fatigue-rated joints	NOT LINKED	Design and verification	UNKNOWN — untested candidate failure mode (Art. 2)

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
BATTERY_FAILURE		design control NOT ESTABLISHED (requires design work)	UNRESOLVED - proposed candidate failure mode (Art. XXV)
adversarial dimension: unsupported_mechanism		design control NOT ESTABLISHED (requires design work)	UNRESOLVED - proposed candidate failure mode (Art. XXV)
adversarial dimension: weak_transfer		design control NOT ESTABLISHED (requires design work)	UNRESOLVED - proposed candidate failure mode (Art. XXV)
adversarial dimension: obvious_combination		design control NOT ESTABLISHED (requires design work)	UNRESOLVED - proposed candidate failure mode (Art. XXV)
adversarial dimension: prior_art		design control NOT ESTABLISHED (requires design work)	UNRESOLVED - proposed candidate failure mode (Art. XXV)
adversarial dimension: contradiction		design control NOT ESTABLISHED (requires design work)	UNRESOLVED - proposed candidate failure mode (Art. XXV)
adversarial dimension: boundary_failure		design control NOT ESTABLISHED (requires design work)	UNRESOLVED - proposed candidate failure mode (Art. XXV)
adversarial dimension: engineering_infeasibility		design control NOT ESTABLISHED (requires design work)	UNRESOLVED - proposed candidate failure mode (Art. XXV)
adversarial dimension: regulatory_incompatibility		design control NOT ESTABLISHED (requires design work)	UNRESOLVED - proposed candidate failure mode (Art. XXV)
link margin collapse at maximum depth/orientation		antenna geometry and pattern; link-margin design with untested electrical loss	UNRESOLVED - proposed candidate failure mode (Art. XXV)
SAR exceedance at transmit power needed for link		transmit power schedule and duty cycle; SAR power compliance and field measurements	UNRESOLVED - proposed candidate failure mode (Art. XXV)
detuning from dielectric	loading of nearby tissue	loading-tolerant antenna topology; characterization across telemedicine	UNRESOLVED - proposed candidate failure mode (Art. XXV)
interference in clinic RF environment		channel selection, filtering, error coding; interference in hospital environment	UNRESOLVED - proposed candidate failure mode (Art. XXV)
antenna feed joint fatigue failure		strain relief at feed; fatigue-rated joint design; NDI verification	UNRESOLVED - proposed candidate failure mode (Art. XXV)

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	record pacing capture and oversensing events before and after thresholding	NOT ESTABLISHED (needs single lead recording)	NOT TESTED	Not applicable as the trial is ongoing
VF-002	run killer experiment: falsification_test_from_killer_device (ERG/calsfc.4457363938049234) PUIS (HFO code=149638000295005)	NOT ESTABLISHED (needs source)	NOT TESTED	
VF-003	link-budget verification in tissue-equivalent phantom	NOT ESTABLISHED (needs source)	NOT TESTED	
VF-004	SAR simulation + measurement against SARfit	NOT ESTABLISHED (needs source)	NOT TESTED	

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED	NOT_PERFORMED

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	biocompatible antenna dielectric	implant telemetry practice	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
biocompatible antenna dielectric	NOT ESTABLISHED	1	biocompatible antenna dielectric	NOT ESTABLISHED	NOT ESTABLISHED	NOT ESTABLISHED	Reliability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- high-density PCB with chip-scale RF front end: domain registry manufacturing route - candidate only; no process qualification exists
- implant-grade antenna integration and feed sealing: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

Source 2:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + canonical invention specification (with epistemic classes)
- + engineering specification (this document, all blocks classed)
- + design input/output graph with explicit IDs
- + engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification)
- + killer-experiment definition and its EIG basis
- + evidence custody references and hashes
- + unresolved-uncertainty register
- + domain equation set (symbolic; sources and applicability judgments identified)

YOU MUST DEVELOP:

- > actual geometry and detailed design
- > sourced engineering parameters and acceptance thresholds
- > physical prototypes
- > all verification and validation results
- > manufacturing process qualification
- > regulatory pathway determination and submissions

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'ALL sourced engineering parameter values', 'reason': 'no measurement or physical observation exists'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	implementing the proposed physical verification	physical verification	NOT ESTABLISHED (to be quoted)	first physical article	NOT ESTABLISHED (to be quoted)
WP-02	constraint-compliance envelope article	per method	constraint-compliance	NOT ESTABLISHED (to be quoted)	constraint-compliance envelope article	NOT ESTABLISHED (to be quoted)
WP-03	article for VF-001	per method	record pacing capture	NOT ESTABLISHED (to be quoted)	article for VF-001	NOT ESTABLISHED (to be quoted)
WP-04	article for VF-002	per method	run killer experiment	NOT ESTABLISHED (to be quoted)	article for VF-002	NOT ESTABLISHED (to be quoted)
WP-05	article for VF-003	per method	link-budget verification	NOT ESTABLISHED (to be quoted)	article for VF-003	NOT ESTABLISHED (to be quoted)
WP-06	article for VF-004	per method	SAR simulation + measurement	NOT ESTABLISHED (to be quoted)	article for VF-004	NOT ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 6-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (pre-registered pass/fail required before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (pre-registered pass/fail required before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.